

TECHNOLOGY



Designing Microsoft Azure Infrastructure Solutions: AZ:305

Design a Governance Solution



Learning Objectives

By the end of this lesson, you will be able to:

- Grasp the principles and importance of design governance to ensure consistency and alignment with organizational objectives
- Apply effective design principles to organize and manage Azure resources within management groups and subscriptions for optimal efficiency and control
- Evaluate landing zones to establish a secure and well-architected foundation for Azure workloads
- Apply resource group design strategies to efficiently organize and manage Azure resources for streamlined administration and maintenance



Learning Objectives

By the end of this lesson, you will be able to:

- Recommend a tagging strategy to categorize and organize Azure resources, facilitating effective resource management and cost tracking
- Evaluate and propose a solution for implementing Azure Policy to enforce compliance and governance standards within an Azure environment
- Apply RBAC (Role-Based Access Control) design principles to ensure secure access management and permissions for Azure resources



Design Governance

Azure Management

Governance is a key aspect of properly Managing Azure resources.

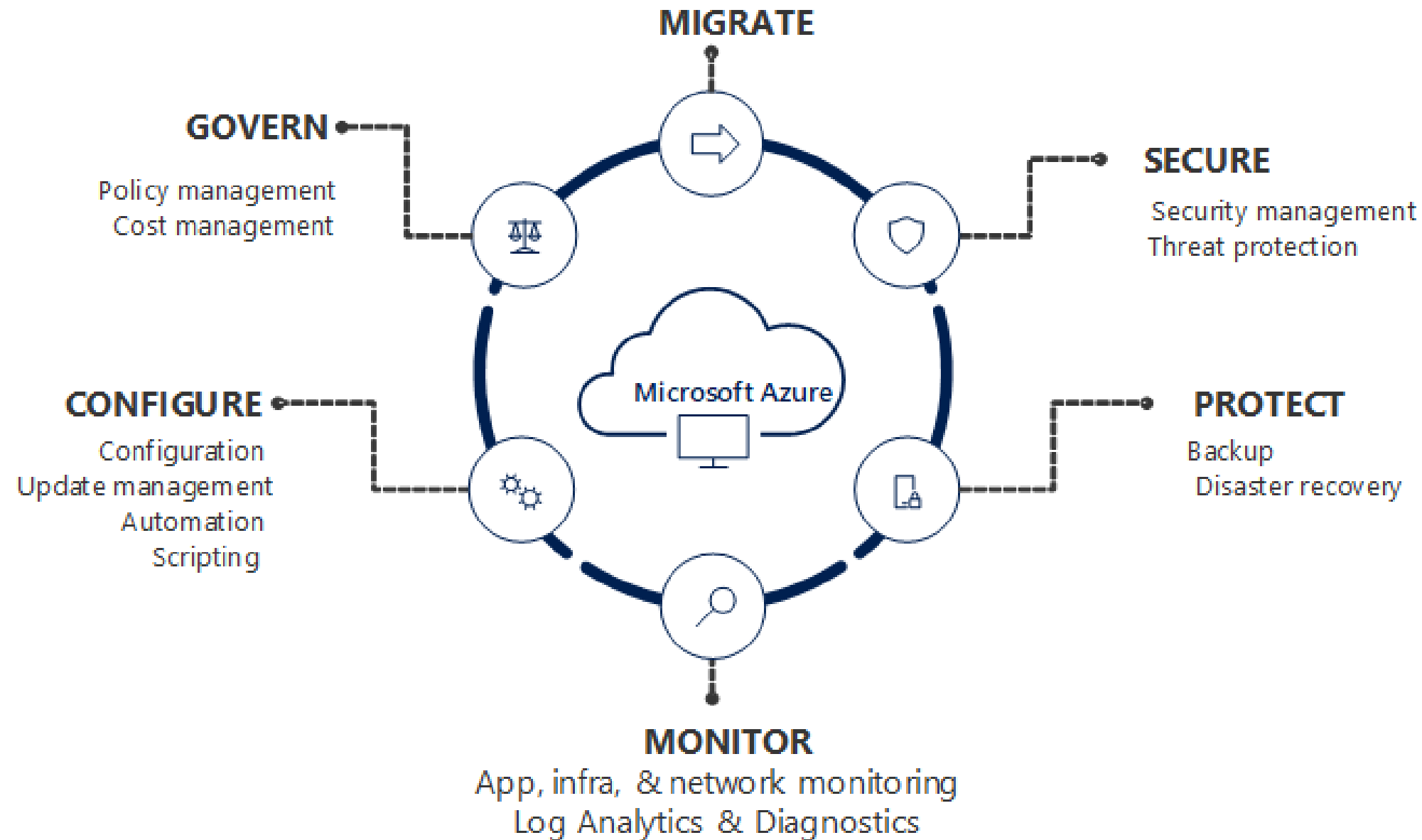
Azure Management is the tasks and processes required to maintain business applications and their supporting resources.

Azure has a suite of services and tools working together to provide comprehensive management.

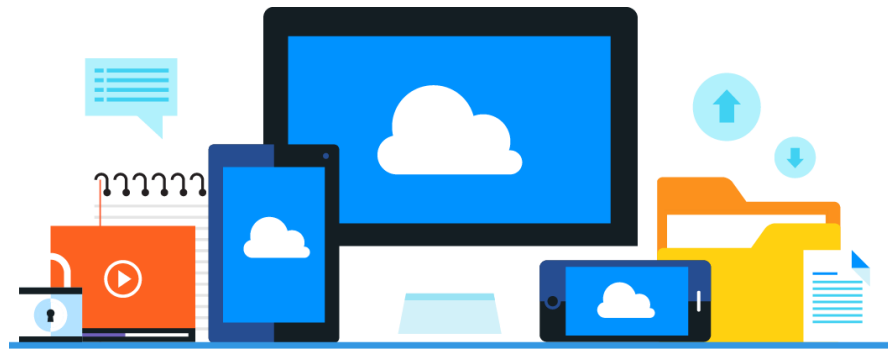


Azure Management

The following are the different components in azure required to manage and maintain any application or resource:



Azure Management



Monitor

It is the act of collecting and analyzing data to audit the performance, health, and availability of resources.



Configure

It refers to the initial deployment and configuration of resources and ongoing maintenance.

Azure Management



Govern

It provides mechanisms and processes to maintain control over applications and resources in Azure.



Secure

It manages the security of resources and data. It involves assessing threats, collecting and analyzing data, and ensuring the compliance of applications and resources.

Azure Management



Protect

It refers to keeping the applications and data available, even with outages that are beyond control.



Migrate

It refers to transitioning workloads currently running on-premises to the Azure Cloud.

Governance Strategies

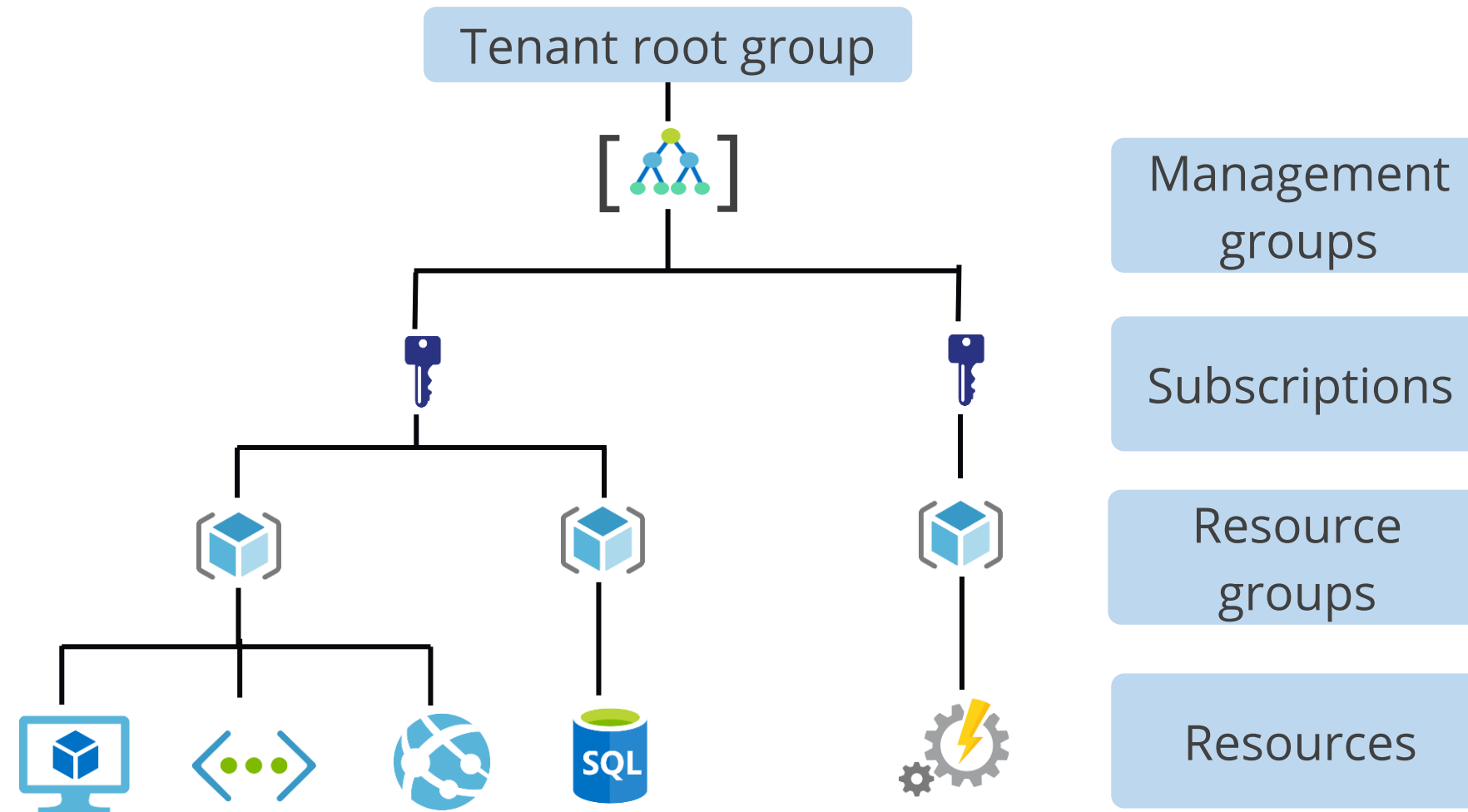
Governance Strategies

- It provides mechanisms and processes to maintain control over the applications and resources in Azure.
- It involves determining the requirements, planning initiatives, and setting strategic priorities.



Governance Strategies

Azure provides a hierarchical structure to apply governance strategies as shown below:



Note

The Tenant root group contains all the management groups and subscriptions.

Governance Strategies

Azure hierarchy has four levels:

- **Management groups** help you manage access, policy, and compliance for multiple subscriptions. All subscriptions in a management group automatically inherit the conditions that are applied to the management group.
- **Subscriptions** logically associate user accounts with the resources that they create. Each subscription has limits or quotas on the number of resources that it can create and use. Organizations can use subscriptions to manage costs and the resources that are created by users, teams, and projects.



Governance Strategies

Azure hierarchy has four levels:

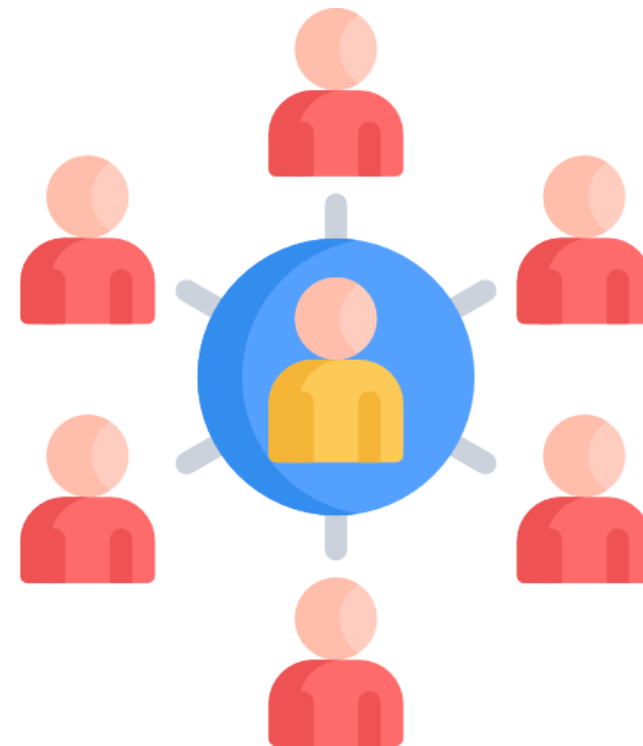
- **Resource groups** are logical containers where you can deploy and manage Azure resources like virtual machines, web apps, databases, and storage accounts.
- **Resources** are instances of services that you can create in a resource group, such as virtual machines, storage, and SQL databases.



Design for Management Groups and Subscriptions

Management Groups

They provide an efficient way to manage access, policies, and compliance across an enterprise.



It manages access through a hierarchy of management groups and subscriptions.



Management Groups

Subscriptions can be organized into containers called management groups.

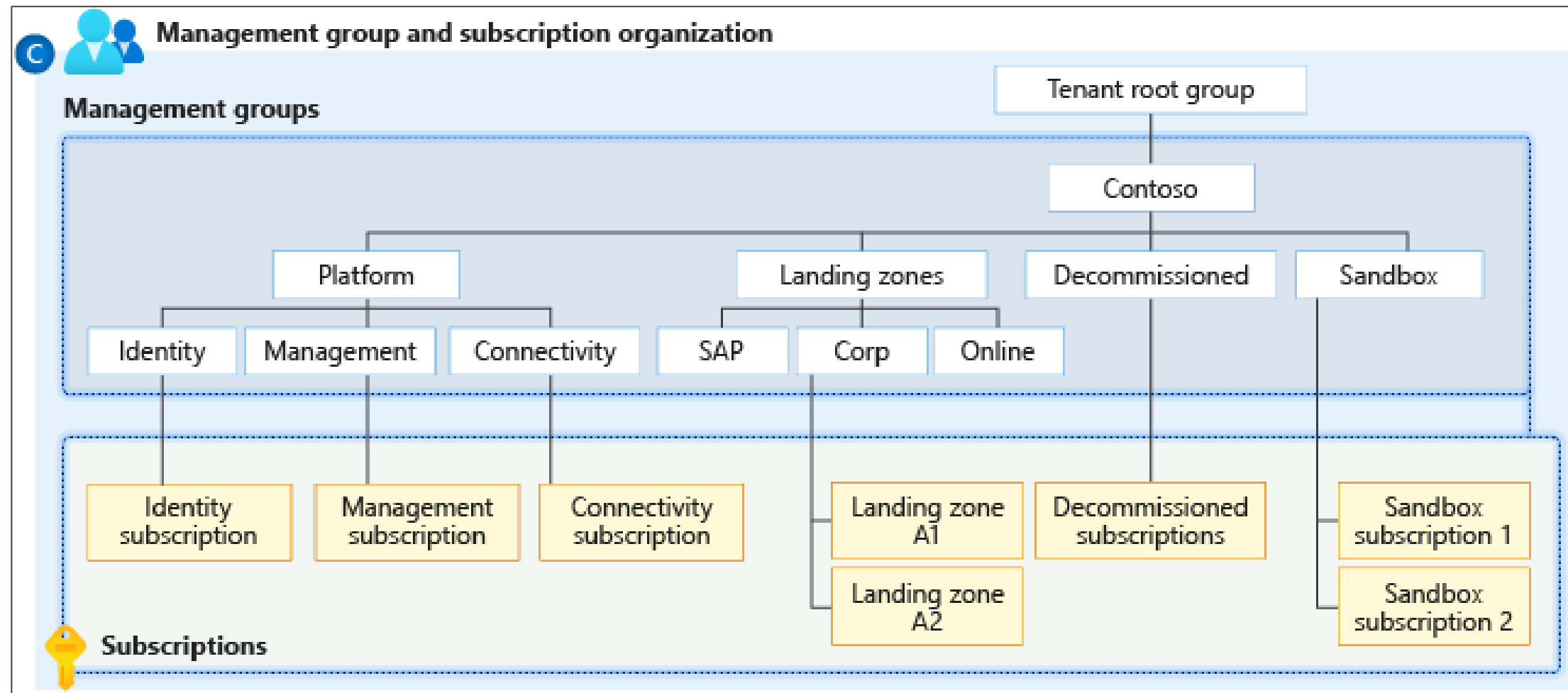
They provide a level of scope above subscriptions.

They are organizationally aligned through custom hierarchies and grouping.

They provide enterprise-level management on a wide scale, regardless of the subscription type.

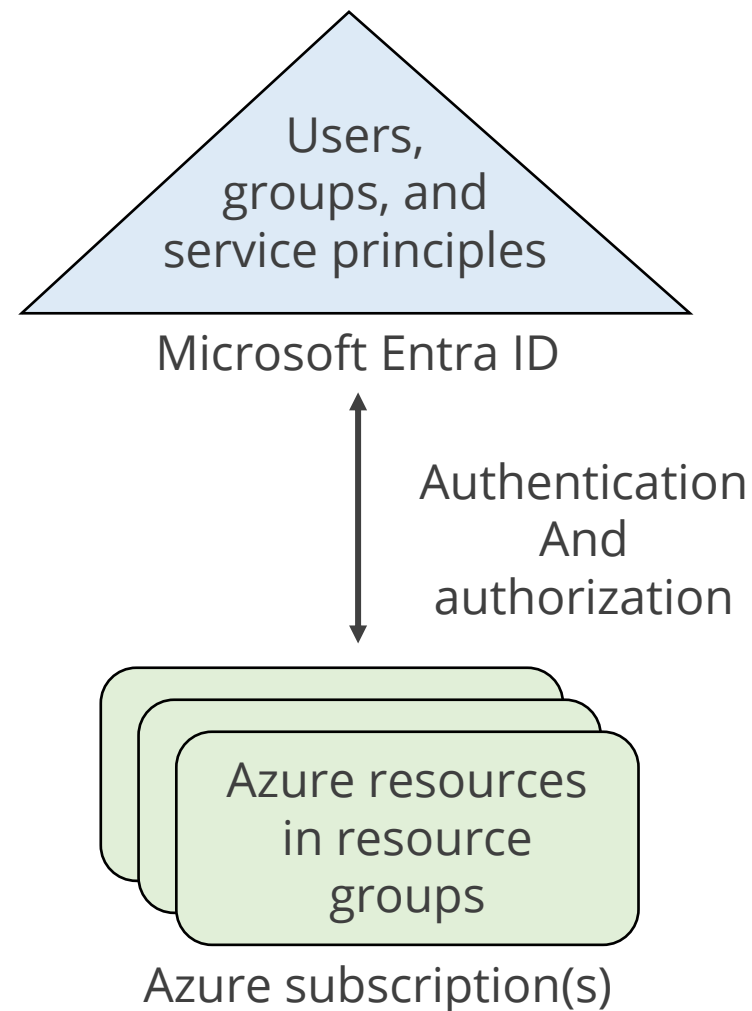
Management Groups

The following diagram shows an example of creating a hierarchy for governance using management groups:



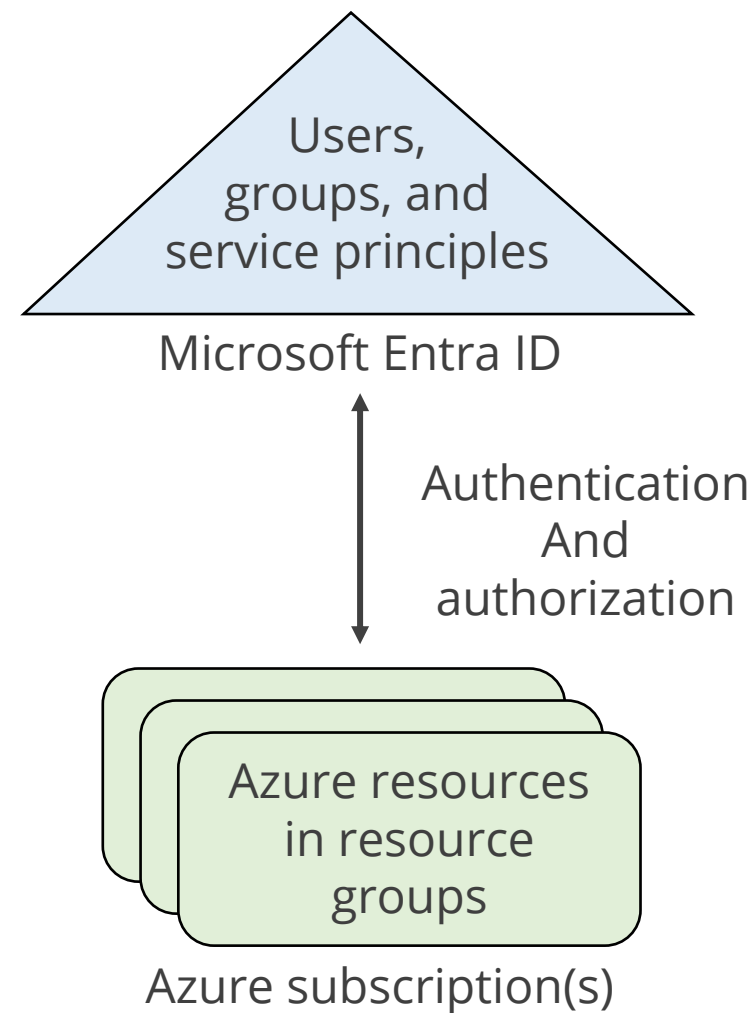
Azure Subscriptions and Accounts

An Azure account is connected to a subscription, which is a logical unit of Azure services.



- Azure services are billed on a per-subscription basis.
- Subscriptions have accounts and are associated with Microsoft Entra ID.

Azure Subscriptions and Accounts



- An account is an identity in Microsoft Entra ID or in a directory that is trusted by Microsoft Entra ID.
- The most common way to allow a user access to Azure services is to add them to the Microsoft Entra ID directory linked to the subscription.

Getting an Azure Subscription

These are the following types of Azure subscriptions:



- **Enterprise Agreement:** The customer makes an upfront monetary commitment to Azure.
- **Reseller:** It is an open licensing program.
- **Microsoft partner:** Find a partner that can design and implement a cloud solution.
- **Free trial account:** Customer can use a free Azure credit to try out different tiers and types of Azure services.

Azure Subscriptions and Service Limits

Microsoft Azure limits are also called quotas.

Managing limits

- Some limits apply to the regional level.
- The user can raise soft limits by raising an online customer support request at no charge.
- These limits keep changing.

Design for Landing Zones

Azure Landing Zone

It provides an infrastructure environment for hosting your workloads.

The significance of a well-designed landing zone includes:

Security

Scalability

Efficiency

Compliance

Azure Landing Zone

It incorporates fundamental principles of governance, security, networking, management, and identity.

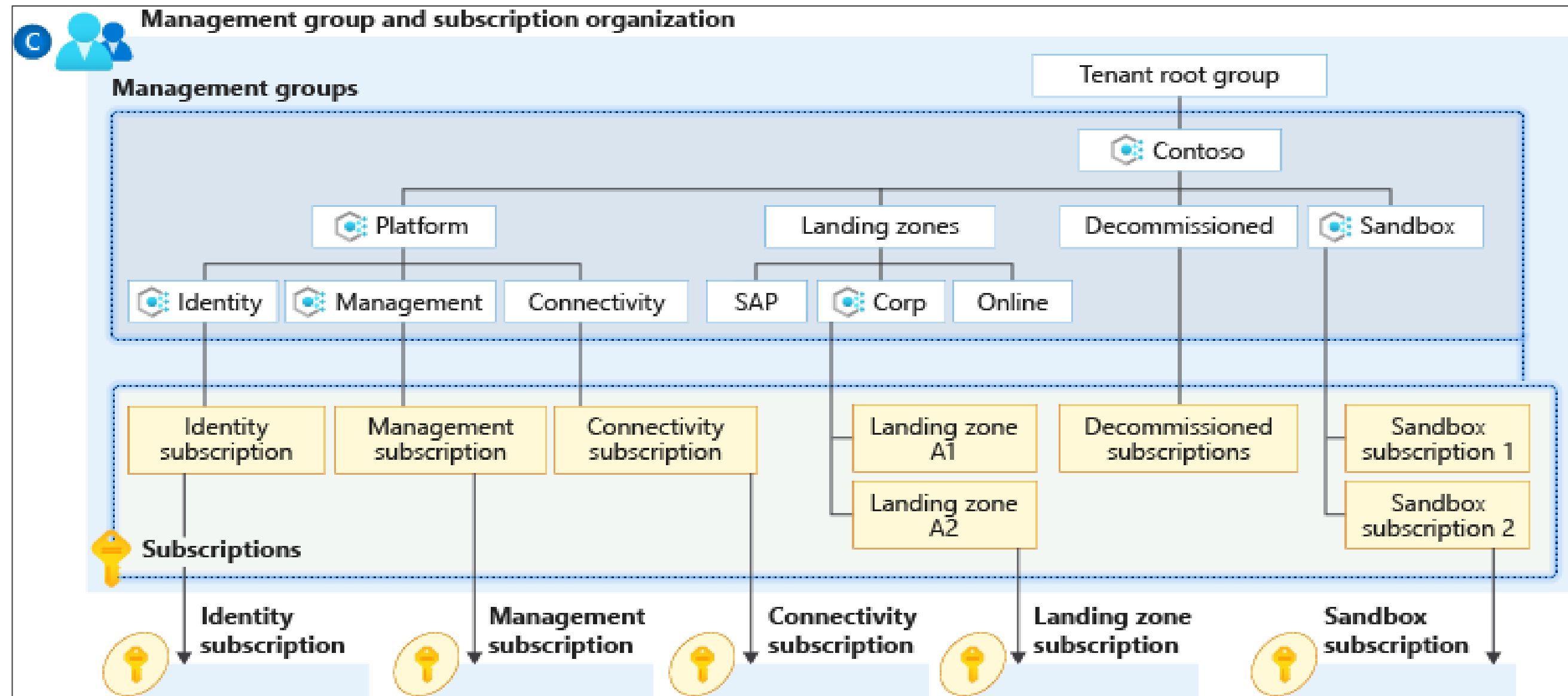
It prepares the environment in advance using code.

It is suitable for both migration projects and new deployments. It allows a seamless transition of existing architectural designs.

It is an integral component of the Cloud Adoption Framework's Ready phase.

Azure Landing Zone: Flow

The following figure depicts the flow of the Azure landing zone:



Azure Landing Zone: Categories

Platform Landing Zone

- A subscription that provides shared services like identity, connectivity, and management for application landing zones.
- Managed by one or more central teams.
- Includes "Identity subscription," "Management subscription," and "Connectivity subscription," each representing different platform landing zones with detailed resources and policies.

Application Landing Zone

- A subscription specifically for hosting applications.
- Pre-provisioned through code and uses management groups for policy control assignment.
- Includes "Landing zone A1 subscription" and "Landing zone A2 subscription," with detailed representation of "Landing zone A2" showing resources and policies.

Azure Landing Zone: Example

Company A

Challenge: Multinational corporation Company A needed to secure its Azure workloads across various regions and services.

Solution: They addressed this by establishing a secure Azure landing zone with integrated controls and policies, ensuring granular access control, compliance, and threat detection.

Outcome: Company A effectively protected its sensitive data, minimized the risk of security breaches, and achieved assurance in terms of regulatory compliance.

Azure Landing Zone: Example

Company B

Challenge: Rapidly growing startup Company B required a solution for efficient management and governance of its expanding Azure resources.

Solution: Implementing a scalable Azure landing zone architecture enabled consistent, automated, and cost-effective provisioning and management of resources.

Outcome: Company B realized substantial cost savings, minimized manual intervention, and streamlined resource scaling as the business expanded.

Azure Landing Zone: Example

Company C

Challenge: Established enterprise Company C sought to modernize its IT infrastructure by migrating on-premises workloads to Azure.

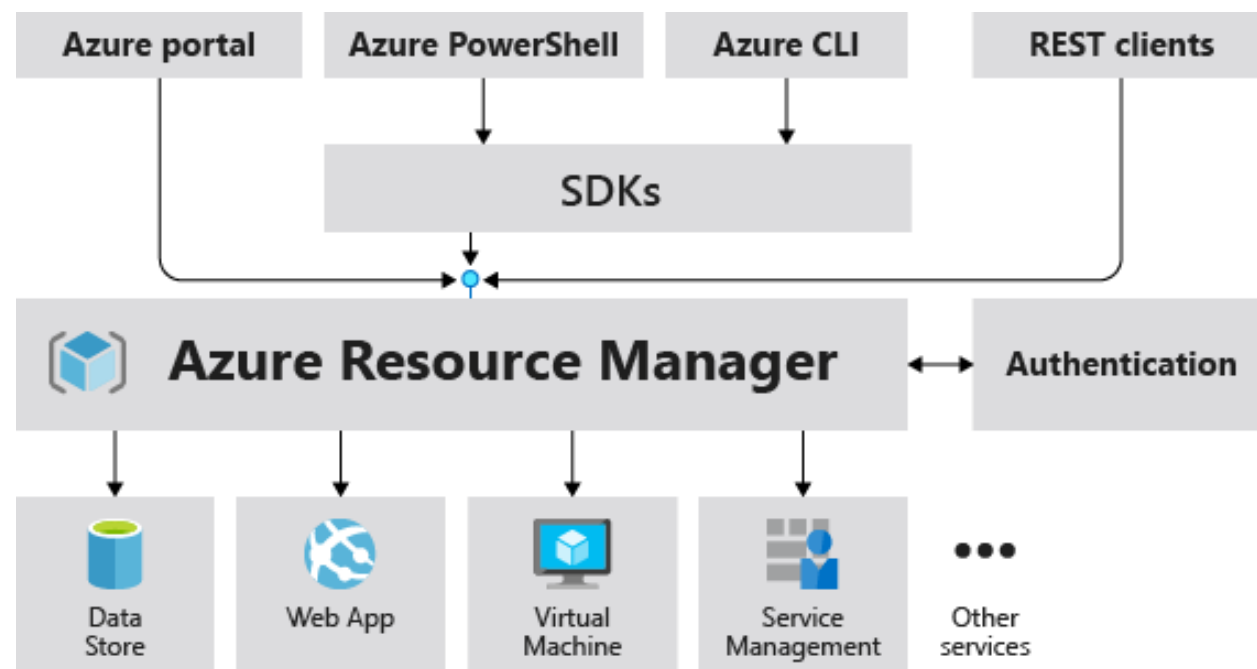
Solution: They implemented a tailored Azure landing zone design for a seamless migration, incorporating existing resources into the new Azure environment.

Outcome: Company C successfully transitioned legacy applications to Azure, gaining cloud-native features and increased agility while maintaining backward compatibility.

Design for Resource Groups

Resource Manager

It is a service that manages and deploys Azure resources.



- Renders a consistent management layer
- Facilitates collaboration with the resources as a group
- Enables deployment, updating, or deletion using a single, coordinated operation
- Provides security, auditing, and tagging features

Resource Groups

It is a logical grouping of resources.



Resource Group

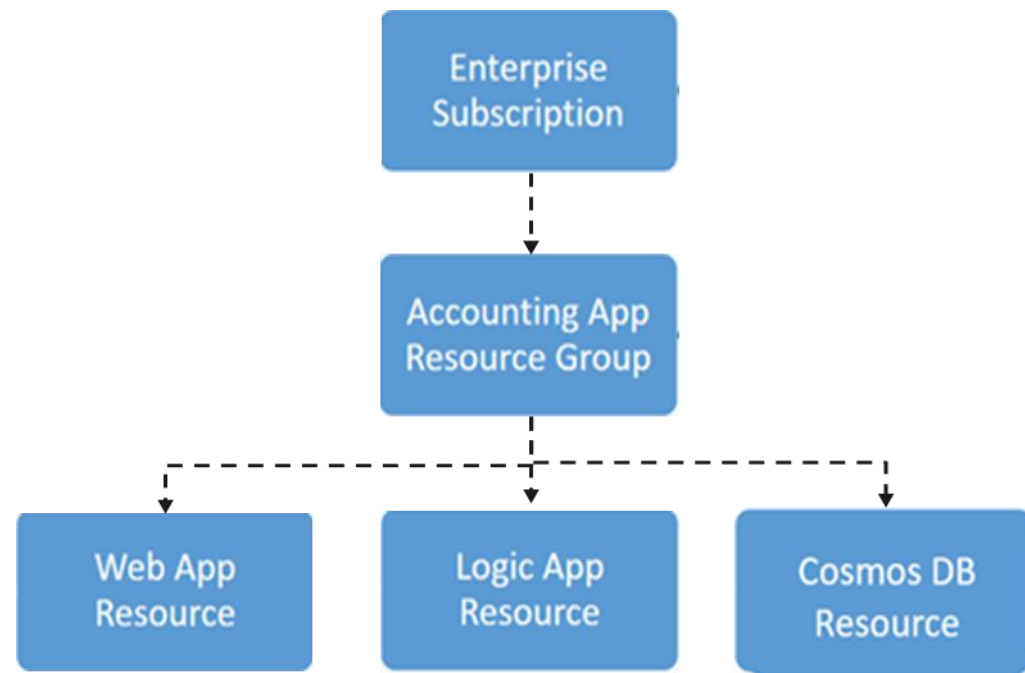


- It is a fundamental concept within the Azure platform.
- It should be closely linked to the lifecycle of the resources.
- A resource must be created within a resource group.
- It must be allocated to each resource.

Note

Most resources can be moved between resource groups.

Resource Groups and Deployments



- Each resource must be in one, and only one, resource group.
- Resource groups cannot be renamed.
- Groups can have a wide range of resources (services). They can have resources from various regions.
- Deployments are made in stages.

Users can easily add, remove, and modify resources by scoping permissions to a resource group.

Resource Group Organization

It is done in the following ways:

For authorization

Since resource groups fall under the scope of RBAC, users can organize resources by who wants to manage them.

For life cycle

Deleting a resource group removes both the group and its contents, making it suitable for disposable resources in production environments.

For billing

When a user places resources in the same resource group, it allows them to be grouped for billing reports.

Recommend a Strategy for Tagging

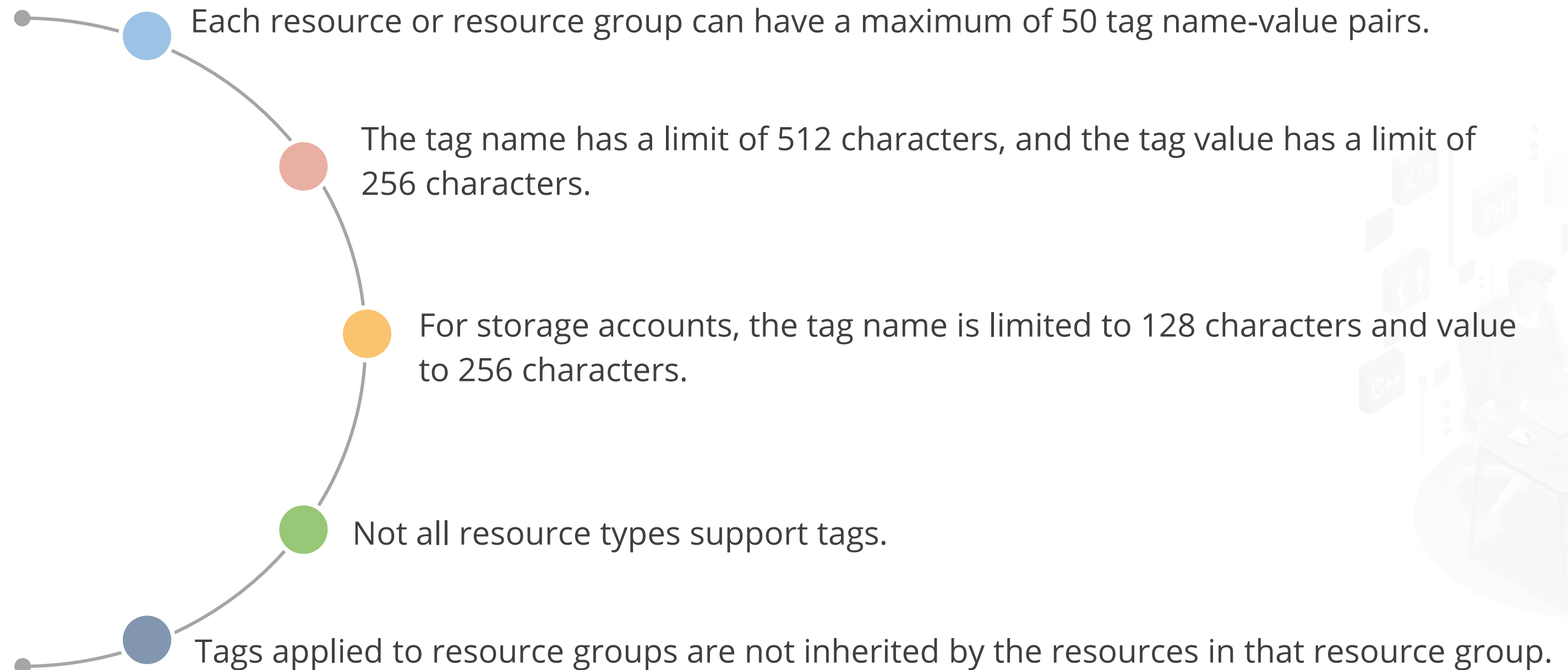
Tags

They logically organize the resources. They consist of a name-value pair and help retrieve related resources from different resource groups.

Daily Usage						
Usage Date	Meter Category	Unit	Consume	Resource Group	Instance Id	Tags
5/14/2015	"Virtual Machines"	"Hours"	3.999984	"computeRG"	"virtualMachines/catalogVM"	"{"costCenter":"finance", "env":"prod"}"
5/14/2015	"Virtual Machines"	"Hours"	3.999984	"businessRG"	"virtualMachines/dataVM"	"{"costCenter":"hr", "env":"test"}"

This approach is helpful when users need to organize resources for billing.

Limitations of Tags

- 
- Each resource or resource group can have a maximum of 50 tag name-value pairs.
 - The tag name has a limit of 512 characters, and the tag value has a limit of 256 characters.
 - For storage accounts, the tag name is limited to 128 characters and value to 256 characters.
 - Not all resource types support tags.
 - Tags applied to resource groups are not inherited by the resources in that resource group.

Tagging: Example

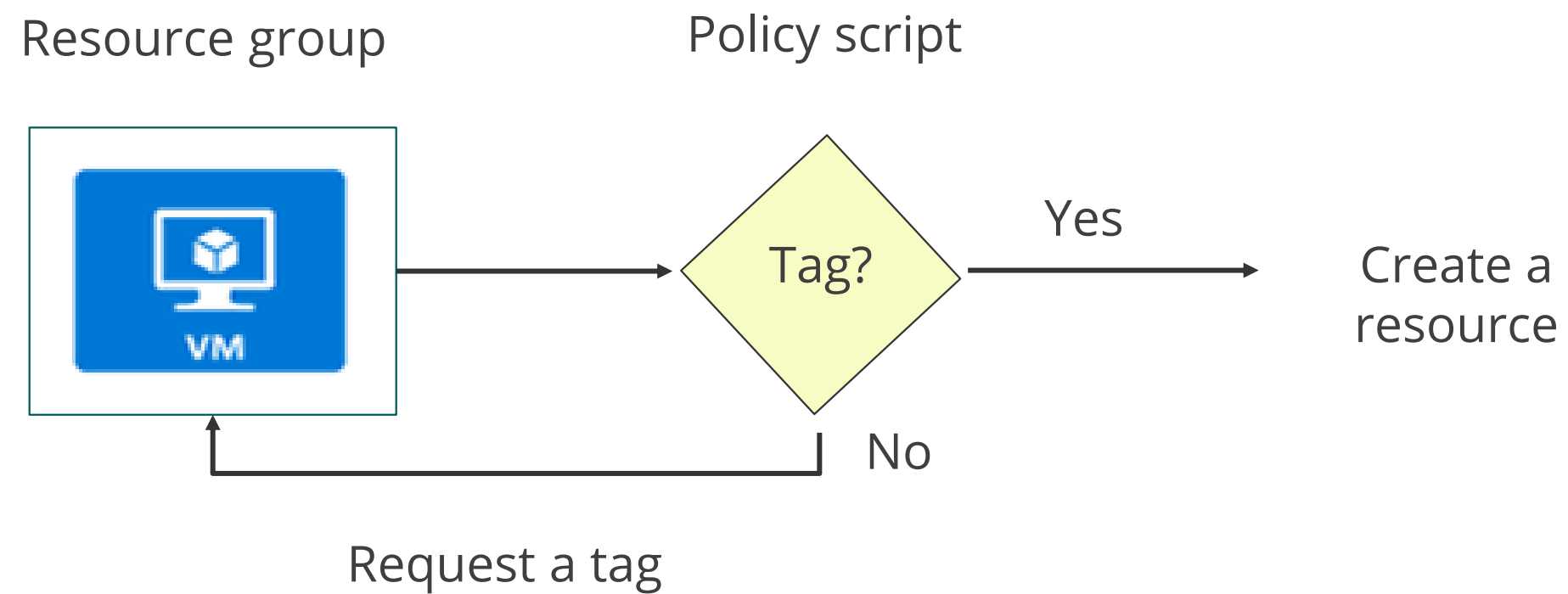
Tag type	Tag	Mandatory/ optional	Description	Data type
Technical	Region	Mandatory	Location	String
Technical	Environment	Mandatory	Dev, Stage, Prod	String
Technical	Maintenance window	Mandatory	Patching Window	String
Automation	Expiration date	Optional	Terminate resource automatically	String
Automation	Time window	Optional	Server online	JSON
Business	Department	Mandatory	Service belonging to department	String

Tagging: Example

Tag type	Tag	Mandatory/ optional	Description	Data type
Business	Application name	Mandatory	Application name	String
Business	Cost center	Mandatory	Cost center ID	String
Business	Description	Optional	Text description of the entity	String
Business	Technical contact	Mandatory	Group responsible for application	String
Security	Data classification	Mandatory	Classification of data	String
Security	Regulatory compliance	Optional	Compliance requirement	JSON

Enforcing Tags with Policy

The workflow of enforcing tags using Azure policy is shown below:



Enforcing Tags with Policy

Policy	Description
Apply tag and its default value	• Appends a specified tag name and value, if that tag is not provided • Specify the tag name and value to apply
Billing tags policy initiative	• Requires specified tag values for cost center and product name • Uses built-in policies to apply and enforce required tags • Specify the required values for the tags
Enforce tag and its value	• Requires a specified tag name and value • Specify the tag name and value to enforce
Enforce tag and its value on resource groups	• Requires a tag and value on a resource group • Specify the required tag name and value

Best Practices: Tagging

Best Practice	Description
Standardize tag names and values	• Use a consistent naming convention (e.g., Environment: Production, Owner: ITDept)
Use tags for resource organization	• Group resources by application, environment, cost center, or business unit
Automate tagging	• Implement policies to enforce mandatory tagging using Azure Policy • Use Azure Resource Manager (ARM) templates, Terraform, or Azure CLI for automated tagging
Monitor tag compliance	• Regularly audit tags to ensure compliance with policies. • Use Azure Cost Management + Billing for tag-based cost analysis.
Limit tag count and length	• Stay within Azure's limits: a maximum of 50 tags per resource, and tag name/value length of 512 characters

Recommend a Solution for Using Azure Policy

Azure Policy

It helps enforce organizational standards and assess compliance at scale.



- The compliance dashboard offers a consolidated view to assess the environment's overall status, allowing detailed analysis at the resource and policy levels.
- It includes implementing governance for resource consistency, regulatory compliance, security, cost, and management.

Example: Users can have the policy to allow only a certain SKU size of virtual machines in your environment.

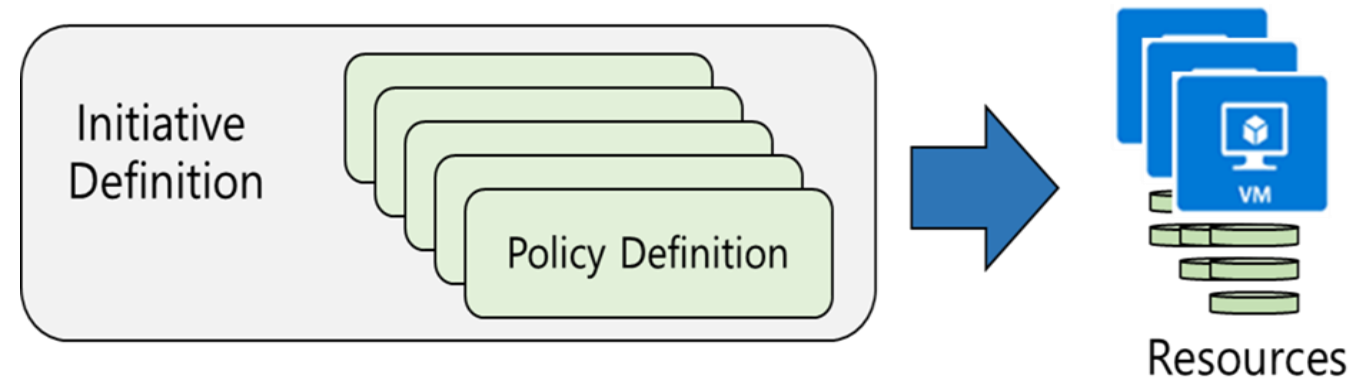
Azure Policy Benefits



- **Enforcement and compliance:** Turn on policies for resources and get real-time policy evaluation and enforcement.
- **Apply policies at scale:** Apply multiple policies and aggregate policy states with the policy initiative.
- **Remediation:** It provides real-time remediation.

Implementing Azure Policies

Following are the steps to implement Azure policies:



- Browse policy definitions
- Create initiative definitions
- Scope the initiative definition
- View policy evaluation results

Browse Policy Definitions

A policy definition defines under what condition a policy is enforced and what effect to take.

Example: Users could prevent VMs from being deployed if they are exposed to a public IP address.



Create Initiative Definitions

Following are the steps to create initiative definitions:

Group policy definition

Define the specific policies that align with your initiative goals within Azure Policy.

Include one or more policies

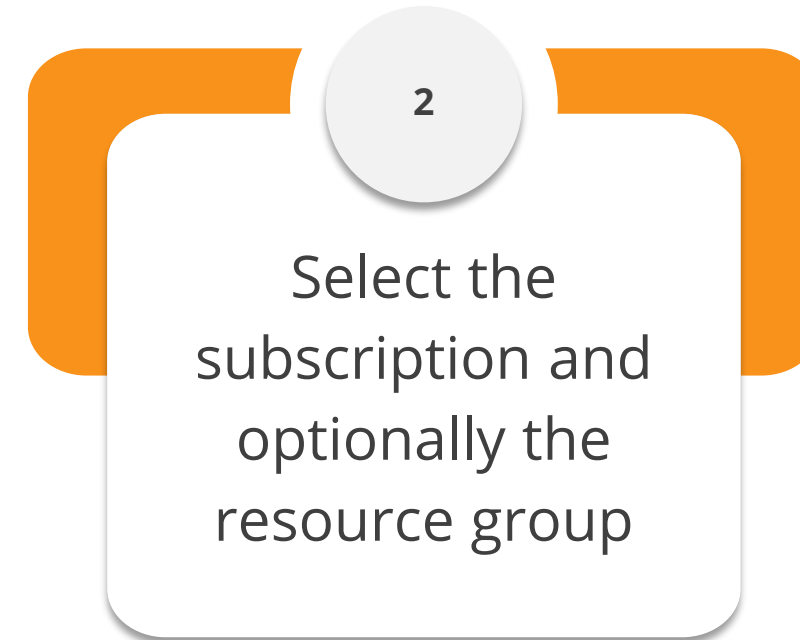
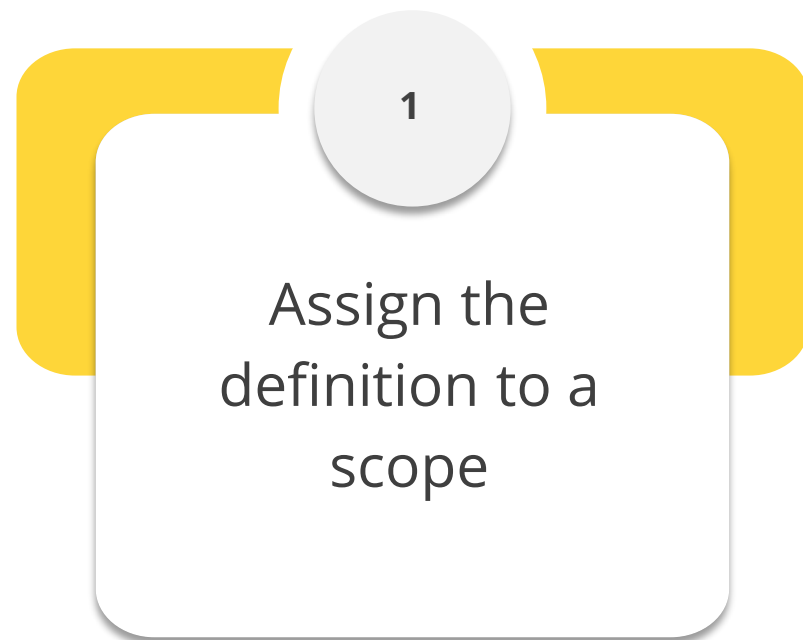
Select and include individual policies that address the desired objectives of your initiative.

Manage planning

Before implementing the initiative, thoroughly plan and strategize to ensure that the selected policies align with organizational goals.

Scope of the Initiative Definition

The scope determines what resources or group of resources the policy enforces. This can be done with the following steps:



Note: An initiative definition can have up to 100 policies.

Policy Effects

The policy creates a list of all assignments that apply to the resource and then evaluates the resource.

Policy Effect	What happens?
Deny	The resource creation/update fails due to policy.
Disabled	The policy rule is ignored (disabled). Often used for testing.
Append	Adds additional parameters/fields to the requested resource during creation or update. A common example is adding tags on resources such as Cost Center or specifying allowed IPs for a storage resource.
Audit, AuditIfNotExists	Creates a warning event in the activity log when evaluating a non-compliant resource, but it doesn't stop the request.
DeployIfNotExists	Executes a template deployment when a specific condition is met. For example, if SQL encryption is enabled on a database, then it can run a template after the DB is created to set it up a specific way.

- Azure policy evaluates the requests to create or update a resource through the Azure resource manager.
- Policy processes several of the effects before handing the request to the appropriate resource provider to avoid violating policy.

Creating Azure Policy



Duration: 10 Min.

Problem Statement:

You have been asked to assist your organization in developing an Azure governance solution that helps enforce corporate standards and analyze compliance at scale as an Azure Architect.

Outcome:

Users will have a robust Azure governance solution that enforces corporate standards, ensures compliance, optimizes costs, and enhances operational efficiency at scale.

Note: Refer to the demo document for detailed steps: 01_Creating_Azure_Policy

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Define a policy



Assigning a Built-in Azure Policy to an Azure Resource Group



Duration: 10 Min.

Problem Statement:

As an Azure Architect, you have been asked to aid your company with an Azure governance solution that can be utilized by Azure Policy to determine which resources are assigned to which policies or initiatives.

Outcome:

By implementing this Azure governance solution, the users will achieve better control over the Azure environment, ensuring resources are managed efficiently and in compliance with organizational standards.

Note: Refer to the demo document for detailed steps:
[02_Assigning_a_Built_in_Azure_Policy_to_an_Azure_Resource_Group](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Assign a policy



Design for Azure RBAC

Azure RBAC

It is a fundamental Azure service that enables fine-grained access control for Azure resources.

It follows the principle of granting permissions based on roles, ensuring the right people have the right access.



Benefits of RBAC

Improved security: Only authorized individuals can carry out specific actions on Azure resources, minimizing the risk of security breaches.

Granular control: It enables detailed management of permissions, advocating a least privilege access model.

Compliance: It helps in meeting regulatory requirements by providing access controls and audit trails.

Efficiency: It streamlines resource management by delegating responsibilities to the right roles.

RBAC: Use Cases

Project-Based Access:

Assign roles based on project teams, limiting access to relevant resources.

Resource Group Level

Control: Use RBAC to control access at the resource group level for a more organized approach.

Compliance and Audit:

Implement RBAC for compliance requirements and auditing needs.

External Collaborators:

Grant restricted access to external collaborators by using custom roles.

RBAC: Best Practices

1

Regular Review:

Periodically review and update RBAC assignments to align with changing roles and responsibilities.

2

Custom Roles:

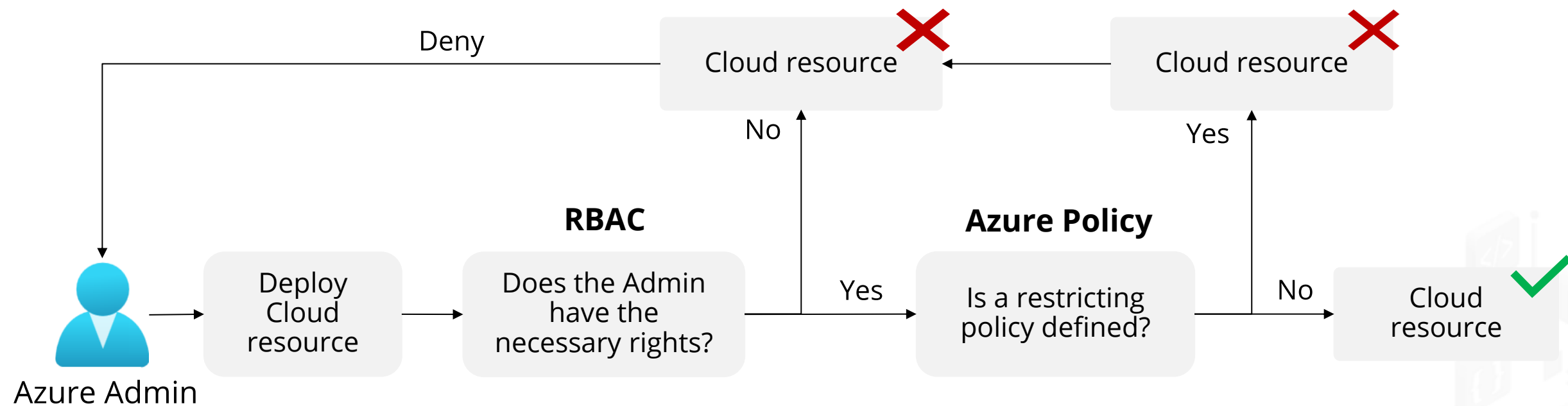
Create custom roles when predefined Azure roles don't match with organization's requirements.

3

Role Segregation:

Avoid over-assigning broad permissions. Segregate duties by assigning minimal necessary permissions.

When to Combine Azure Policy and Azure RBAC ?



RBAC examples

- Does the Admin have the right to deploy?
- Does the Admin have the right to deploy this resource type?
- Does the Admin have the right to deploy this resource group?

Policy examples

- Is the region restricted?
- Is the resource type restricted?
- Should a tag be applied?

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Case Study

Design a Governance Solution

Scenario

ABC is planning on making a few significant changes to its governance solution. They have asked for your assistance with recommendations and questions. Here are the specific requirements:

- **Cost and accounting:** ABC has two main business units that handle apparel and sporting goods. Each of the business units consists of three departments: product development, marketing, and sales. Each business unit and subunit will be responsible for tracking their Azure spend. At the same time, the enterprise IT team will be responsible for providing company-wide Azure cost reporting.

Design a Governance Solution

Scenario

- **New development project:** The company has a new development project for customer feedback. The CFO wants to ensure all costs associated with the project are captured. For the testing phase, workloads should be hosted on lower-cost virtual machines. The virtual machines should be named to indicate they are part of the project. Any instances of non-compliance with resource consistency rules should be automatically identified.

Design a Governance Solution

Requirements

Cost and accounting:

- What are the different ways ABC could organize their subscriptions and management groups? What approach would be the most effective in meeting their requirements?
- Design two alternative hierarchies and explain your decision-making process.

Design a Governance Solution

Requirements

New development project:

- What are the different ways ABC could track costs for the new development project?
- How are you ensuring compliance with the requirements for virtual machine sizing and naming?
- Propose at least two ways of meeting the requirements. Explain your final decision.

How are you incorporating the Azure well-architected framework pillars to produce a high-quality, stable, and efficient cloud architecture?

Key Takeaways

- Design governance ensures consistency, compliance, and alignment with organizational goals in Azure architecture.
- Effective design of management groups and subscriptions optimizes resource organization and simplifies administration.
- Landing zone design establishes a secure and well-architected foundation for deploying Azure workloads.
- Azure RBAC is a fundamental Azure service that enables fine-grained access control for Azure resources.





Knowledge Check

**Knowledge
Check**

01

Which of the following elements in the Azure management hierarchy are instances of services that users create?

- A. Resources
- B. Management groups
- C. Resource groups
- D. Subscriptions



**Knowledge
Check**

01

Which of the following elements in the Azure management hierarchy are instances of services that users create?

- A. Resources
- B. Management groups
- C. Resource groups
- D. Subscriptions



The correct answer is **A**

Resources are instances of services that users create.

On what basis are Azure services billed?

- A. Per subscription
- B. Enterprise agreement
- C. Browse policy
- D. Storage



Knowledge Check

02

On what basis are Azure services billed?

- A. Per subscription
- B. Enterprise agreement
- C. Browse policy
- D. Storage



The correct answer is **A**

Azure services are billed on a per-subscription basis.

What are the advantages of Azure policies?

- A. Enforcement and compliance
- B. Subscription
- C. Remediation
- D. Both A and C



Knowledge Check

03

What are the advantages of Azure policies?

- A. Enforcement and compliance
- B. Subscription
- C. Remediation
- D. Both A and C



The correct answer is **D**

Enforcement, compliance, and remediation are the advantages of Azure policies.

Which of the following is NOT true with respect to tags?

- A. Tags help in organizing resources for billing or management.
- B. Tags applied to resource groups are inherited by the resources in that resource group.
- C. Tags consist of a name and a value.
- D. Each resource or resource group can have a maximum of 50 tag names or value pairs.



Knowledge Check

04

Which of the following is NOT true with respect to tags?

- A. Tags help in organizing resources for billing or management.
- B. Tags applied to resource groups are inherited by the resources in that resource group.
- C. Tags consist of a name and a value.
- D. Each resource or resource group can have a maximum of 50 tag names or value pairs.



The correct answer is **B**

Tags applied to resource groups are not inherited by the resources in that resource group.

Design Governance

Duration: 20 min.



Project Agenda: Design a solution for governance with respect to the given scenario

Description: You work as a cloud architect in a Fortune 500 organization. You need to design a solution for developers that will grant them the ability to provision certain Azure Resources, keeping the below requirements in mind:

- Allow only a certain size of VMs to be provisioned
- Allow storage account and VM provisioning in only specific regions
- Do not allow the creation of a storage account if a secure transfer is not enabled

Design Governance

Duration: 20 min.



Perform the following:

1. Create users
2. Create a group
3. Create a resource group to assign policy as per the requirement
4. Assign a role to the created group at the resource group level
5. Assign the policies to resource group

LESSON-END PROJECT

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Thank You